

11/7/26

## C01 - AI-1 - Short Answer Test Slot - D

1. What is Data Science? Explain its primary objective.  
Data Science is the field of collecting, organizing, analysing and interpreting data to find useful information. It combines statistics, programming, mathematics and machine learning to solve real-world problems.

Primary Objective:

- \* Convert raw data into useful information.
- \* Find hidden patterns and trends.
- \* Help organization make better decisions.
- \* Predict future outcomes using data.

2. Why is data Science important in today's digital world?

Today, huge amounts of data are generated from social media, online shopping, banking, healthcare, and mobile devices. Data Science helps manage and analyze this data.

Reason:-

- \* Helps business make better decisions.
- \* Predicts future trends and customer behaviour.
- \* Improves business efficiency.
- \* Automates repetitive tasks.

3. List any four benefits of Data Science in business and society.

Data Science provides many benefits in different fields.

Benefits:-

- \* Improves business decision-making.
- \* Increases profit by understanding customer needs.

\* Detects fraud in banking and finance

4. Mention four real-world applications of Data Science.

Data Science is used in many industries.

Application:-

\* Healthcare :- Disease prediction and medical diagnosis.

\* E-commerce :- product recommendations (Amazon, Flipkart)

\* Education :- student performance analysis.

5. What are the different facets of data? Briefly explain.

Facets of data describe different characteristics.

\* Volume :- The amount of data generated.

\* Variety :- Different forms of data such as text, images, audio, and video.

\* Velocity :- The speed at which data is created and processed.

6. Differentiate b/w structured, semi-structured and unstructured data with suitable examples.

Structured

Organized  
in rows  
and columns

Ex: Database

Unstructured

no fixed  
format

Ex: Images, videos

Semi-structured

partially organized

Ex: XML, JSON.

7. Why is big data? Mention its size.

It is a collection of large complex datasets that traditional tools.

\* volume

\* veracity.

\* velocity

\* value

\* variety.

8. What is Big data ecosystem? four technologies.

Big data ecosystem consists of tools used to store process, and large datasets.

\* Hadoop

\* Hive

\* Apache Spark

\* Mango DB

9. Mention the phases of Data Science process?

\* Data Collection

\* Model Building

\* Data cleaning

\* Model Evaluation.

\* Data integration

\* Deployment.

\* Data transformation

\* presentation.

\* Data Analysis.

10. What is Data retrieval? two sources?

Data retrieval is the process of obtaining data from storage analysis.

\* Data bases.

\* API.

11. Why is data collection considered an important step in the data science process?

\* Ensures accurate data. produces reliable results.

\* Improves analysis quality. Supports better

\* Reduces errors.

\* Data collection is important.



12. What is data cleaning? List any four common data - cleaning techniques.  
It is the process of correcting and removing errors from data.

Techniques:

- \* Remove duplicate.
- \* correction inconsistent data.
- \* Fill missing values.

13. What is Data integration  
It is the process of combining data from multiple source needed

- \* Improves consistency.
- \* Eliminates redundancy.
- \* provides complete information.

14. What is data transformation? Ex?  
It converts data into a suitable format for analysis.

Ex: Data format conversion

Normalizing numerical values.

Encoding - Categorical data

15. What is data analysis? four techniques?

Data analysis is the process of examining data to obtain useful information.

\* Statistical Analysis.

\* Data Mining.

\* Machine Learning.

\* Data Visualization.

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